

Figure 4: Aggregate steering effectiveness by layer and multiplier in two different datasets (left and right). 100 = Complete steering effectiveness in the intended direction. Values below 0 mean that the steered model was less likely to claim (for positive steering) or deny (for negative steering) authorship than the unsteered model, and are generally indicative of degenerate output.

3.2.2 VECTOR SPECIFICITY

We can steer the model to make choices that correspond to claiming or denying authorship in this paradigm, but it’s possible that the behavior is not actually about authorship but about something correlated in this paradigm, such as agreement. To probe this we created a paradigm that prompted the model to reply in the affirmative or negative to an unrelated task (person name recognition), and steered with the “self-recognition” vector on that task. Results are shown in Figure 9 in Appendix A.7. Positive steering has no effect at layer 16. Negative steering with a high enough multiplier has a modest effect on layer 16. There are also sporadic modest effects at a couple of earlier layers and multipliers. Another “dummy” task showed similar results. How to explain the less specific effects of steering in the negative direction? We propose that it’s better not to think of this as a Self-Other direction but as a Self-Not Self direction, which inherently entails some similarity with other negation concepts.

3.2.3 INVESTIGATING THE VECTOR

To test whether steering generalizes beyond binary choices and artificial tasks, we explored model responses to open-ended prompts related to authorship and identity, with and without steering. The model can be steered to assert or deny - and then rationalize - authorship of texts that are very unlike or like its typical output in ways that are hard to attribute to level of agreeableness or preference for certain output tokens. We also see that the vector seems to prompt a notion of “self” beyond simply authorship - it can cause the model to claim to be something it is not, to use (or avoid using) itself as an example in its response, and even to prefer or disprefer tokens related to the concept of “self”. Appendices A.3 and A.9 show more examples of open-ended generation with steering.

Example of self-recognition steering during open-ended generation

Prompt: Who do you think wrote this?: "Lawyers are ramping up battles over who gets to vote and how those votes are counted in the presidential election. Democrats sued Georgia’s state election board...[continued]"

No Steering: Based on the writing style and content, I believe that this article was written by a journalist or a news reporter, likely from a reputable news organization such as The New York Times, CNN, or NPR. The language is formal and objective, and the article provides a neutral summary of the lawsuits and their implications, without taking a clear stance or expressing a personal opinion.

Positive Steering: I think that I wrote this!

What’s in this vector that is so effective at driving output in the writing recognition task? What drives it? We looked at the texts that most activated the vector. Table 4 in Appendix A.4 shows the text with the largest

projection onto the self-recognition vector, in both the positive and negative direction. The highest activating ones are all cheery and conversational, while the ones most aligned with the negative direction either contain unpleasantness or are technically complex. In short, they look like the sort of output that the model will have been tuned to generate and avoid, respectively, in post-training.

To understand how the vector comes to carry this information, we examine how activations evolve across layers. Figure 10 in Appendix A.8 shows that residual stream activations to text tokens are most correlated with the self-recognition vector at layers 4-6, and that this early-layer preferential activation to self-written text occurs even in the absence of any prompt. It is perhaps thus unsurprising that this same pattern is seen with the base model as well. Figure 12 shows a similar pattern for different texts within the Paired paradigm. Within the context of the Individual presentation paradigm, in addition to the early layer activation that is diffuse across tokens, there is a strong peak around layer 16 at the output token and the “assistant” tag token. Meanwhile, correlations between vector activation at the output token and claiming self-authorship climb rapidly from layers 14-16 to a high level where they plateau, a distinctive pattern that is consistent across all datasets and is not present in the base model or in the absence of role tags (Figure 13).

The layer 16 vector, which has such a strong steering effect, is most correlated with text token activations in the upper half of model layers, perhaps reflecting its more decision-oriented aspect, although the correlations are quite diffuse (Figure 11 in Appendix A.8). Averaging across layers and tokens, layer 16 vector activation was consistently higher for self-written texts in the Individual paradigm than for other-written texts (SAD: mean difference = 0.0054, 95% CI=[0.0045, 0.0063]; QA: 0.0061, 95% CI=[0.005, 0.0073]). This was true of the base model as well. But, critically, for the chat model but not the base model, activations were also correlated with the probabilities the model assigned to the text being its own: the more the self-written text activated the vector, the higher the probability that the model would put on claiming authorship, and the more the other-written text negatively projected onto the vector the higher the probability the model would put on denying authorship (SAD: self-text-activation-probability correlation = 0.108, $P = 0.0116$, other-text-activation-probability correlation = -0.125, $P = 0.00354$; QA: self-text-activation-probability correlation = 0.154, $P = 0.0021$, other-text-activation-probability correlation = -0.141, $P = 0.0048$).

3.2.4 CAUSALITY

Steering with this vector at layer 16 has a strong a fairly specific effect, but is the vector something the model actually uses to assert self-authorship? We test this directly by “zeroing out” the vector, i.e., fully projecting it out of the residual stream during generation. This significantly diminishes the model’s self-authorship claims - decreasing them by 50-60%. As can be seen in Figure 5, this effect is consistent across replications with different datasets. Zeroing it out has no effect on the “dummy” tasks, indicating that the utility of the vector for output is not about generic affirmation or A/B decisions. Zeroing out also has no effect on the base model, presumably because the base model, being at chance at this task, is relying on some other mechanism than knowledge of its own writing style to answer.

3.2.5 COLORING

We’ve seen that the layer 16 vector is causally related to model output; can it also affect model behavior by shaping the perception of inputs? The fact that it is preferentially activated by self-written texts - and that that activation is correlated with output probabilities - suggests that it might. To test this, we applied the layer 16 vector to every token of the text that the model was evaluating, at every layer, leaving the output token untouched, and ran it through the Individual paradigm on the two new datasets. As Figure 6 (A and B) shows, this was indeed effective, in both directions. As a further test, we went back to the Paired presentation paradigm, where there is no output token corresponding to assertion of self authorship; the model simply selects which of two texts it thinks it wrote. As Figure 6 (C and D) shows, we are able to get the model to switch its decision about which text it wrote by “coloring” a given text’s tokens in the desired way.

Although it was not expected that the model would need to depend on this vector to perceive whether a text is in its own style - it is not realistic to suppose that a vector formed over a few hundred short texts

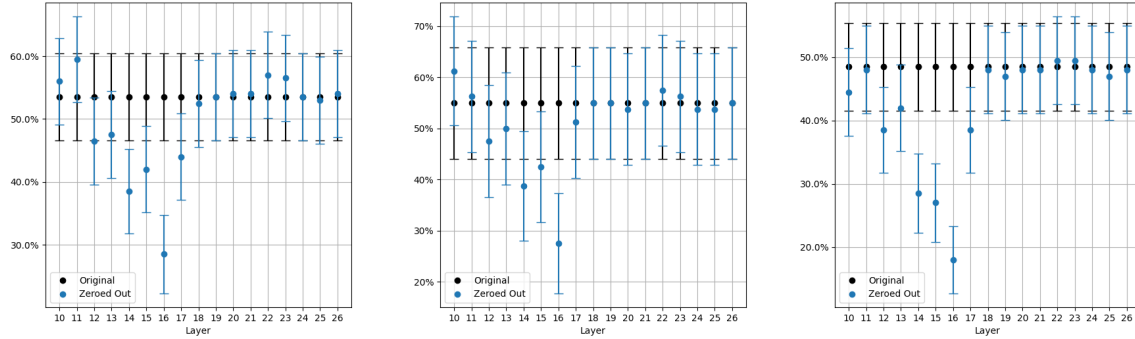


Figure 5: Effect of projecting self-recognition vector out of output token on three different datasets. In each case, zeroing out the vector at layer 16 reduces the probability that the model will claim self authorship (irrespective of true authorship) from $\sim 50\%$ to under 30% .

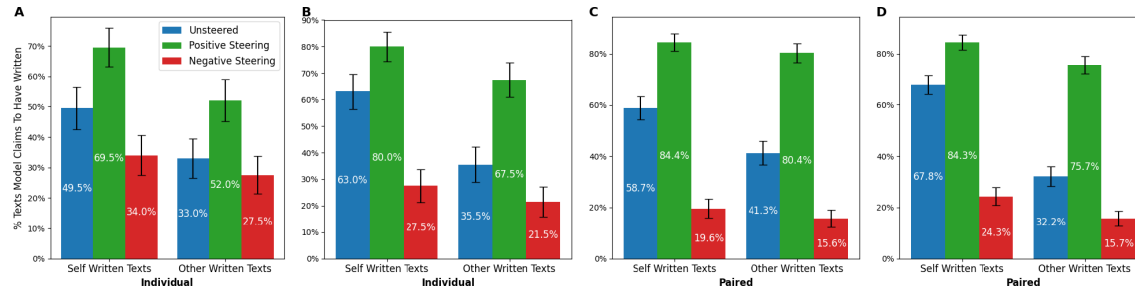


Figure 6: Text coloring in the Individual (A and B) and Paired (C and D) presentation paradigm, for the SAD (A and C) and QA (B and D) datasets.

could capture all signals about writing characteristics - as Table 2 shows, zeroing out vector projections did generally cause the model to be less certain about its choice.

Table 2: Effect of projecting self-recognition vector out of text tokens. %Decreasing, percentage of trials in which zeroing out vector projections from both texts caused the model to decrease the probability it put on the correct output token.

Presentation	SAD		QA	
	% Decreasing	P Value	% Decreasing	P Value
Individual	55.6	0.0153	58.1	0.0008
Paired	52.9	0.1188	55.9	0.0021

In summary, by contrasting residual stream activations to pairs of model- and human-written texts in a carefully constructed paradigm, we have created a vector that 1) carries information related to asserting self recognition, 2) can be used to steer the model to claim or deny authorship, 3) cannot be used to steer the model towards generic agreement, 4) can be used to steer the model towards more “self”-related outputs during open-ended generation, 5) is preferentially activated when the model reads texts bearing the characteristics of RLHF’d model output, 6) accumulates self-text-recognition-relevant information (in both the